

MISP Introduction and recent new features

CIRCL - 2026

 <https://www.misp-project.org>

July 13, 2026

MISP Project <https://misp-project.org>

Introduction

Workshop Goals

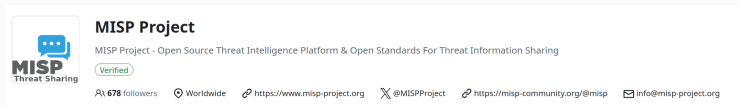
Audience

- **Primary:** Cyber Threat Intelligence Analysts
- **Secondary:** SOC and Incident Response Teams
- **Expected MISP Knowledge:** None
- **Training level:** Basic

Workshop Objectives

- Understand how MISP supports **operational investigations**
- Transform unstructured data **actionable intelligence**
- **Correlate** incidents across teams and partners
- **Enrich** data and **Collaborate**

- **Presentation of MISP (1h)**
 - Introduction (30min)
 - Basic UI, Concept & Terminology (30min)
- **Break (15min)**
- **Demonstration and Familiarization (1h30)**
 - Walkthrough on encoding data (30min)
 - Faster encoding & Enrichment (20min)
 - Getting The Most Out Of Your Data (20min)
 - Collaboration and Sharing (20min)
- **Lunch Break (1h)**
- **Practical Activity (30min): Ransomware infection**
- **Brief on Global MISP instance (1h)**



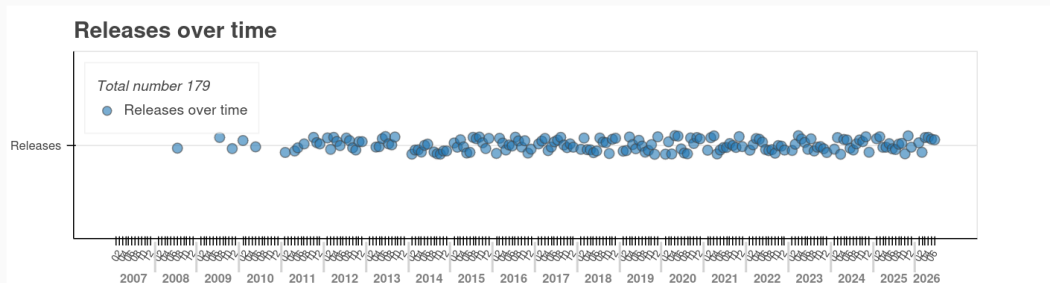
- CIRCL is mandated by the Ministry of the Economy (under NIS 2).
- CIRCL leads the development of many OSS.
- **CIRCL manages multiple large MISP communities**, enabling active daily threat intelligence sharing.
- Funding comes from Luxembourg, various EU programs, and partnerships under EU/US agreements.

Introduction

MISP Introduction

What is MISP?

- MISP¹ is a **threat intelligence sharing platform (TISP)** that is FREE & OPEN SOURCE software
- Mature project that started in 2012, and has been following a **community-driven development** since then



¹<https://www.misp-project.org/>

What is MISP?

- Used worldwide to **manage and share threat-related information and intelligence**.
- **Open-Source Commitment:** Users of MISP can rely on the tool remaining open source and never becoming closed source (autonomy of the users).



Number recorded on March 2026

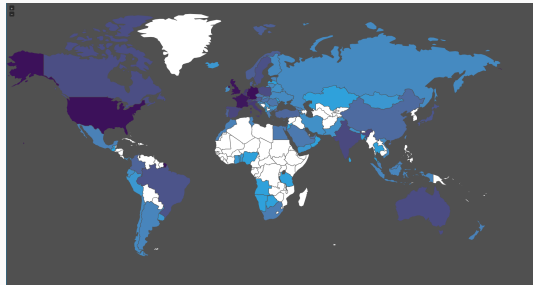
What is MISP?

- A tool that **collects** from partners, your analysts, your tools, feeds, ...
- Normalises, **correlates**, **enriches** the data
- Allows teams and communities to **collaborate** and **share**
- **Feeds** automated protective tools and **disseminate**

Who is Using MISP?

Communities: Groups of users sharing within a set of common objectives and values.

- **Private Sector**
Financial, telecom, manufacturing
- **Military Organizations**
Military and governmental CERT
- **Security Vendors**
Vendor-led intelligence sharing
- **ISACs**
- **Trusted Groups**
Inc. air-gapped trusted sharing
- **Law Enforcement Agencies**
- **International Groups**



Aggregated from misppriv.circl.lu (June 2026)

MISP and Threat Intelligence

MISP is designed from the ground up to perform **context-rich threat intelligence**

- **Enrich** information with context and metadata
- **Maps Threats and TTPs**
e.g MITRE ATT&CK
- Supports many **standardized classification** marking
- Enables information **curation** through automated quality checks
- Generates customizable **threat reports**
- Allows creation of **Dashboard** for trend analysis

ATT&CK heatmap

Events / technique 0 100

Initial Access	Execution	Persistence	Privilege Escalation	Credential Access	Discovery
Phishing T1566 182	Command and scripting interpreter T1059 463	Boot or Login Autostart Execution T1547 13	Process Injection T1055 18	Input Capture T1056 148	System Information Discovery T1002 23
Exploit Public Facing Application T1190 12	User Execution T1054 83	Scheduled Task/job T1053 9	Host or Login Autostart Execution T1547 13	Credentials from Password Store T1056 42	Process Discovery T1002 16
Nonpersistent Link T1182 18	Malicious File T1034 302 95	Valid Accounts T1078 8	Registry Run Keys / Subkeys T1547 100	Unstructured Credentials T1056 2	Unstructured Credential Exposure T1400 18
Valid Accounts T1078 8	Malicious Link T1034 301 5	Modify Registry T1132 3	Structural Modification T1547 109 1	Brute Force T1110 1	File and Directory Discovery T1005 15
Spearphishing Attachment T1199 1	Customs Library T1034 105 1	Create or Modify System Objects T1043 3	Scheduled Task/job T1053 9	Credentials from Web Services T1056 3	Query Registry T1112 13
Drive-by Compromise T1189 4	Scheduled Task/job T1053 9	External Remote Services T1133 2	Valid Accounts T1078 8	Credentials in Files T1001 1	Account Discovery T1002 12
Quality Chain Compromise T1195 4	Native API T1138 7	Server Software Component T1069 2	Corruption for Privilege Escalation T1199 4	Modify Authentication Process T1056 1	System Owner/User Discovery T1002 3
External Remote Services T1133 2	Windows Management Instrumentation T1047 5	Host Execution Flow T1074 2	Access Token Manipulation T1134 4	OS Credential Dumping T1003 1	Software Discovery T1002 4
Trusted Relationships T1109 2	Exploitation for Client Execution T1035 5	BTFS Jams T1187 1	Create or Modify System Processes T1043 3	Private Keys T1140 1	System Network Configuration T1002 7
	Shared Modules T1129 4	Create Account T1136 1	Drive Execution Control Mechanism T1040 2	Steal Web Session Cookies T1009 1	System Location Discovery T1004 4
	Mailbox T1170 1	U.S. Service Order Hijacking T1038 1	Host Execution Flow T1074 2		Application Window Discovery T1002 4
	Scanning T1084 1	Event Triggered Execution T1540 1	OS Search Order Hijacking T1038 1		Security Software Discovery T1002 3
	System Services T1100 1	External Modules and T1025 4	Event Triggered Execution T1040 2		System Service Discovery T1002 3

MITRE ATT&CK Matrix

MISP provides mechanisms to maintain, enrich and operationalize intelligence throughout its lifecycle.

✓ Data Quality

- False Positive Management & Feedback Loops
- Knowledge Base and Vocabularies
 - ▷ *Address the Who, What, How and Why.*

✚ Enrichment

- Enrichment system
- Multiple Correlation types
 - Raw, Fuzzy Hash, CDIR

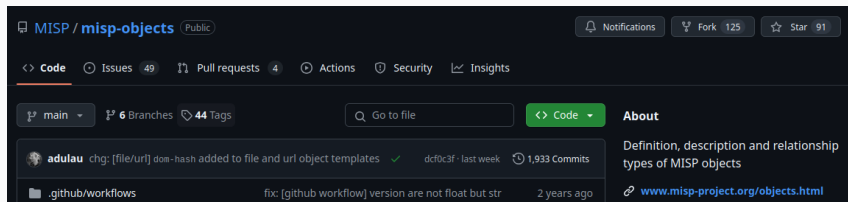
🔑 Governance

- Review Workflows
- Publication Control
- Lifecycle Management

🔗 Integration

- API & Automation ecosystem
- Integration with External Tools
- Notebooks

Flexible Data Format



MISP Objects provide reusable templates to model virtually any investigative artifact.



Network

IPs, URLs, hashes



Devices

Phones, software



Payments

Cards, transactions, crypto



Vehicles

Cars, aircraft, drones



Identities

Persons, passports, PNR



Extensible...

420+ Templates

- **Typical CTI**

- Threat Actors
- Countries
- Attack Techniques
- Campaigns
- And **many more** ...

- **Impact**

- Geographical / Sectorial
- Demographical
- Economical / Legal / Regulatory
- Assets / Users
- Affected availabilities / Data Integrity

- **Confidentiality**

- Copyrighted / Classified
- Disclosure

- **Responsibility**

- Governance - 3rd party Hosted
- Management / Ownership
- Confidence / Estimative Language
- Discovery

- **Extensible** ...

- 350+ Different Vocabularies

Using the Power of the Community

MISP provides multiple mechanisms to foster collaboration between analysts, teams and communities.



Collaboration

- Proposals
- Sightings
- Extended Events



Discussion

- Analyst Notes
- Opinions



Specialized Sharing

- Delegation
- Sharing Groups
- Trust Communities

New features in the past few months

MISP Highlights

New features in the past few months - June 2026

16

Releases

2851

Total Commits

167

Pull Requests

72

Contributors

1619 commits on MISP core • 1232 commits across 18 modules • 145 merged pull requests

- **UI/UX modernisation**
- New **dashboard** system
- New **event templating** system
- **Geolocation & map** visualisation
- **Continuous Security Hardening** — Regular updates remain critical

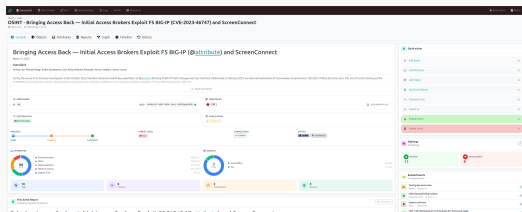
The biggest visual change in years!

- **Theme system** — opt-in themes let users preview new UIs
 - **BetaUI** — a community-contributed UX rework (Chris Horsley, Cosive)
A redesigned, responsive event index & reorganised navigation
 - **Overmind** — the next-generation interface
Refreshed look and new views
- **Decomposed event view** — the monolithic event view split into specialized views
- **Modern front-end stack** Migrating MISP to **Bootstrap 5** and **Font Awesome 7**

UI/UX modernisation

The screenshot shows a form titled "Add Event" with a subtitle "Describe the threat event in precise terms...". The form includes a "Create event" button, a "Description" field, a "Severity" dropdown menu, a "Status" dropdown menu, a "Priority" dropdown menu, a "Category" dropdown menu, a "Tags" field, and a "Create event" button. The form is designed to be user-friendly and intuitive.

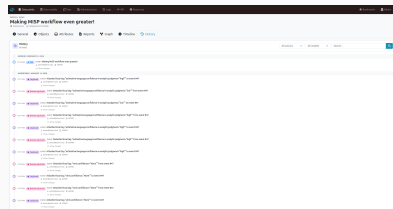
/event/add



/event/view

The screenshot shows a configuration page for a REST client. It includes a "REST Client" section with a "Name" field, a "URL" field, a "Method" dropdown menu, a "Headers" field, a "Body" field, and a "Save" button. The page also features a "Filtering" section with a "Filter" dropdown menu and a "Filter" field.

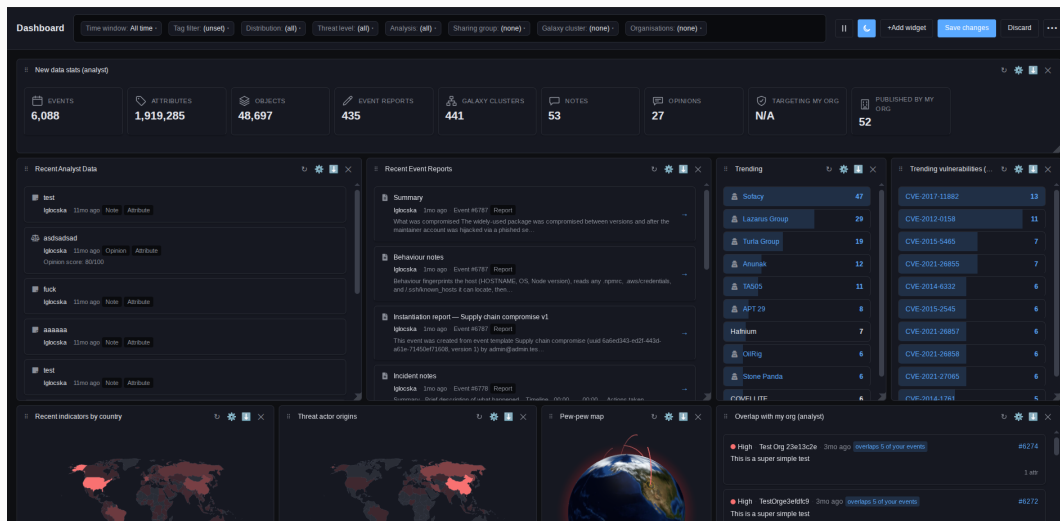
/api/rest



/event/view-history

- A redesign of the MISP dashboard
- **Widget gallery** for discovering and adding widgets, with on-demand thumbnails
- Richer **chart-based widgets**
- Full UI support of parameters and **global dashboard filters**
- Foundation for **analyst-focused dashboards** and per-organisation views

New dashboard system



- A **total rewrite** of MISP's templating engine
- **Drag-and-drop builder UI** with inline validation, object-template / taxonomy / galaxy pickers, and a live preview
- A **Templates library** ships ten ready-to-use starter templates (ransomware, credential exposure, spearphishing, supply-chain, ...)
- Migration helper for legacy templates, and full documentation

New dashboard system: Templates Library & Migration helper

Search

Search by name, UUID, or description

	ID	Name	UUID	Organisation	Distribution	Active	Sections	Elements	Source	Version	Modified
	#1903	Spearphishing email triage	11d82511-c161-4154-8f06-b01bc8527c69	Iglocska		X	5	12	Library	v1	Last Modified 2026-04-29 08:39:58
	#1904	Supply chain compromise	6a6ed343-ed2f-443d-a61e-71450ef71608	Iglocska		X	6	18	Library	v1	Last Modified 2026-04-29 08:39:58
	#1905	Suspicious domain triage	420ebb51-952e-4059-9550-af95090434aa	Iglocska		X	5	12	Library	v1	Last Modified 2026-04-29 08:39:58
	#1906	UAV observation	d33ba254-707a-4eb7-a168-c8bb8c3b1730	Iglocska		X	5	13	Library	v1	Last Modified 2026-04-29 08:39:58
	#1907	Vulnerability disclosure	ae855b4a-146c-44eb-88e7-a531d76f431d	Iglocska		X	5	13	Library	v1	Last Modified 2026-04-29 08:39:58
	#1898	Credential exposure	f04246eb-c26a-4d69-a602-885ab7f8b38b	Iglocska		X	6	16	Library	v1	Last Modified 2026-04-29 08:39:57
	#1899	Espionage case triage	294bf4d1-c2b2-47ed-94d4-07ef5a3730e3	Iglocska		X	4	17	Library	v1	Last Modified 2026-04-29 08:39:57
	#1900	Financial fraud case	d938ef21-bb49-46b8-9f37-e7af9da0b7ba	Iglocska		X	5	19	Library	v1	Last Modified 2026-04-29 08:39:57
	#1901	Malware sample submission	3685adb8-dd42-4835-b5b9-dae131d25583	Iglocska		X	6	15	Library	v1	Last Modified 2026-04-29 08:39:57
	#1902	Ransomware incident	47011a02-2069-4d07-8bf1-e0fd93e635a	Iglocska		X	6	16	Library	v1	Last Modified 2026-04-29 08:39:57

New dashboard system: User form

UAV observation

Capture a UAV/drone observation for civil safety monitoring, military situational awareness, or intelligence collection. Includes drone platform details, sighting location, observed behavior, and optional attribution.

☒ Step-by-step

UAV platform

UAV platform details using the dedicated MISP `uav` object template.

Observed UAV ^{*} misc · v4

Populate the most reliable platform identifiers first (manufacturer, model, serial number). Add mission clues when known.

At least one of: `model`, `manufacturer`, `serial-number`

 Expand to fill required — empty

 Add another instance

[← Previous](#)

Step 2 of 5

[Next →](#)

New dashboard system: Builder UI

Edit Event Template



Library-managed template

This template is managed by the `misp-event-templates` submodule. The next *Update from library* run will overwrite your edits unless you uncheck *Library-managed* below — that flips this row to a fork and library updates will leave it alone.

Name

UAV observation

Distribution

Community

☐ Active

☒ Library-managed

Description (Markdown)

Capture a UAV/drone observation for civil safety monitoring, military situational awareness, or intelligence collection. Includes drone platform details, sighting location, observed behavior, and optional attribution.

Add element

Section

Text block

Attribute field

Object field

Tag field

Galaxy field

File field

Event report

Object reference

SECTION Observation overview

s_overview

ATTRIBUTE Observation context — Internal reference/text

observation_context

ATTRIBUTE Date observed — Other/datetime

date_observed

ATTRIBUTE Short summary — Other/comment

summary

SECTION UAV platform

s_platform

OBJECT Observed UAV — uav v4

obj_uav

SECTION Observation location

s_location

OBJECT Sighting geolocation — geolocation v8

obj_geo

Attribute field properties

Stable id

observation_context

Label *

Observation context

Help text (Markdown)

Choose the operational framing for this report. Suggested values: 'civil', 'military', 'intelligence'.

☒ Mandatory (user must fill this field)

☐ Repeatable (user can add multiple values)

MISP category *

Internal reference

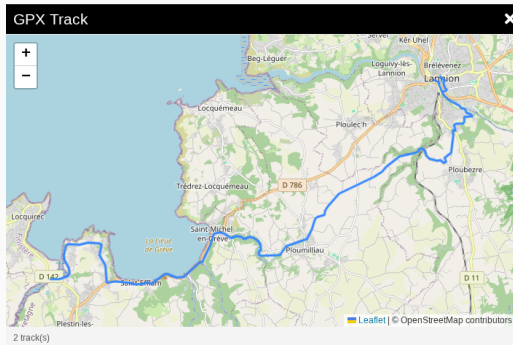
MISP type *

- Objects with geographic coordinates are now rendered on **interactive tile-based maps**
- **Accuracy-radius** visualisation
- Native **GeoJSON** rendering for complex shapes and features
- **GPX track** support for tracking objects

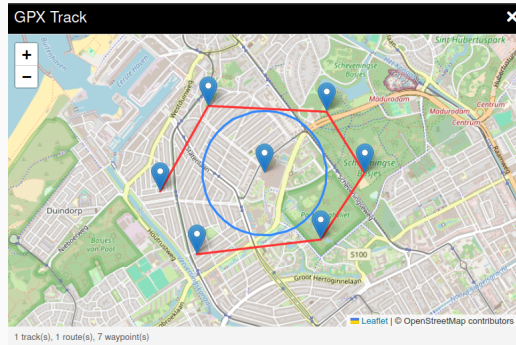
Disabled by default

configurable tile server (defaults to `geo.circl.lu`)

Geolocation & map visualisation



Full track



Routes and waypoints